Hands-on: The DFT+U method

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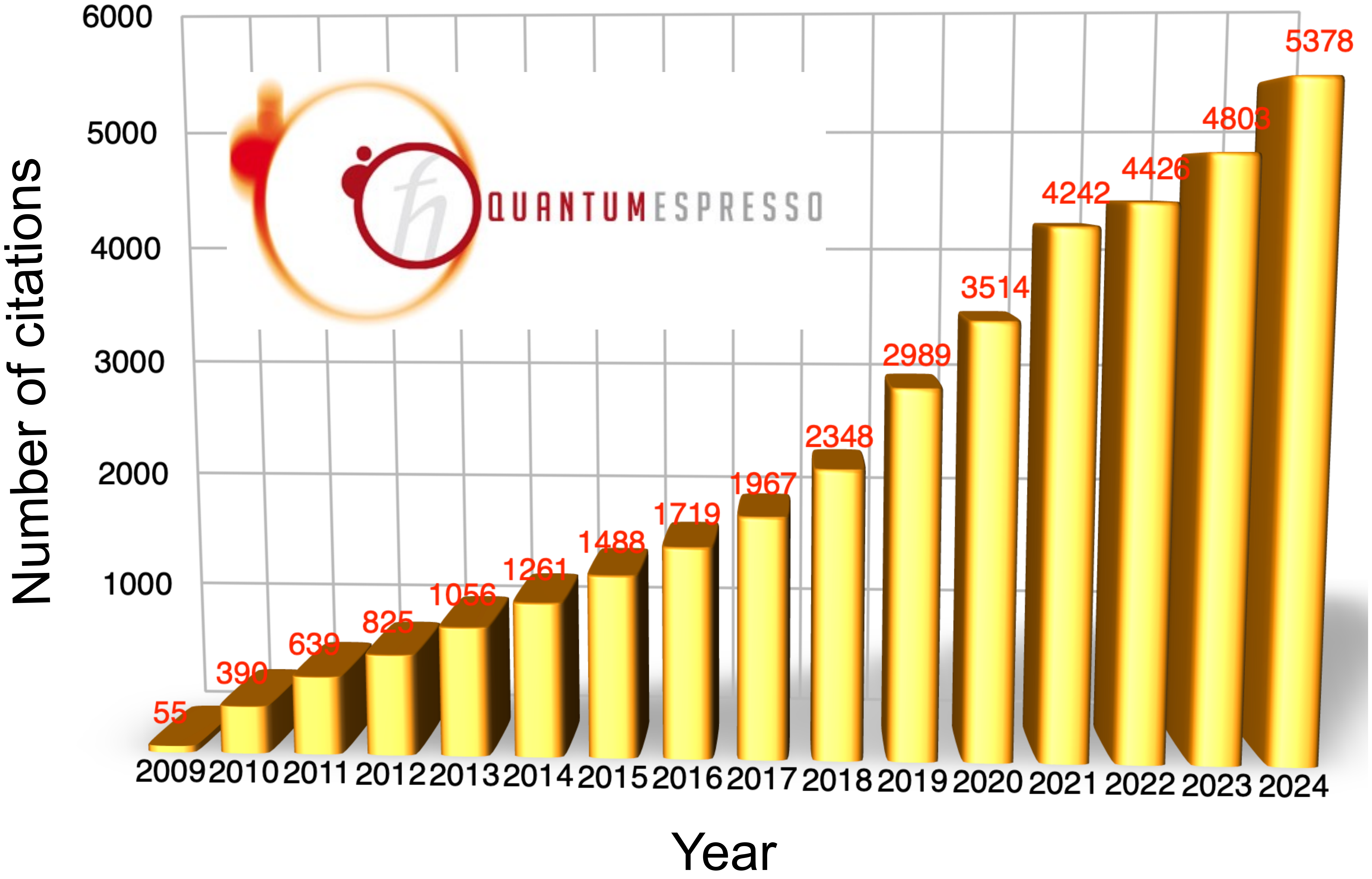
PSI ETH4D

ASESMA

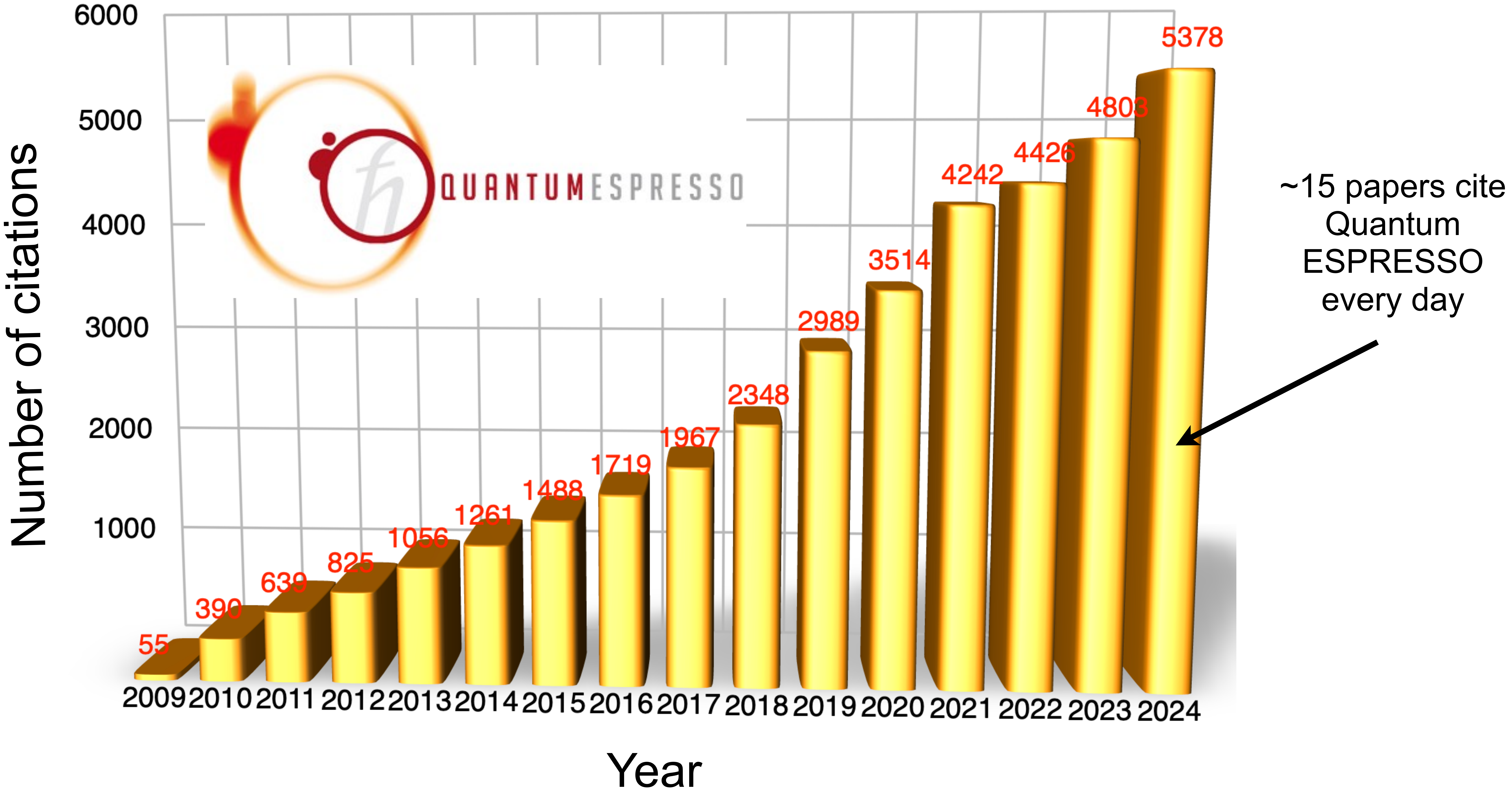

open-source
QUANTUM ESPRESSO
ELECTRONIC-STRUCTURE CODE
FOR MATERIALS SCIENCE APPLICATIONS

Quantum ESPRESSO



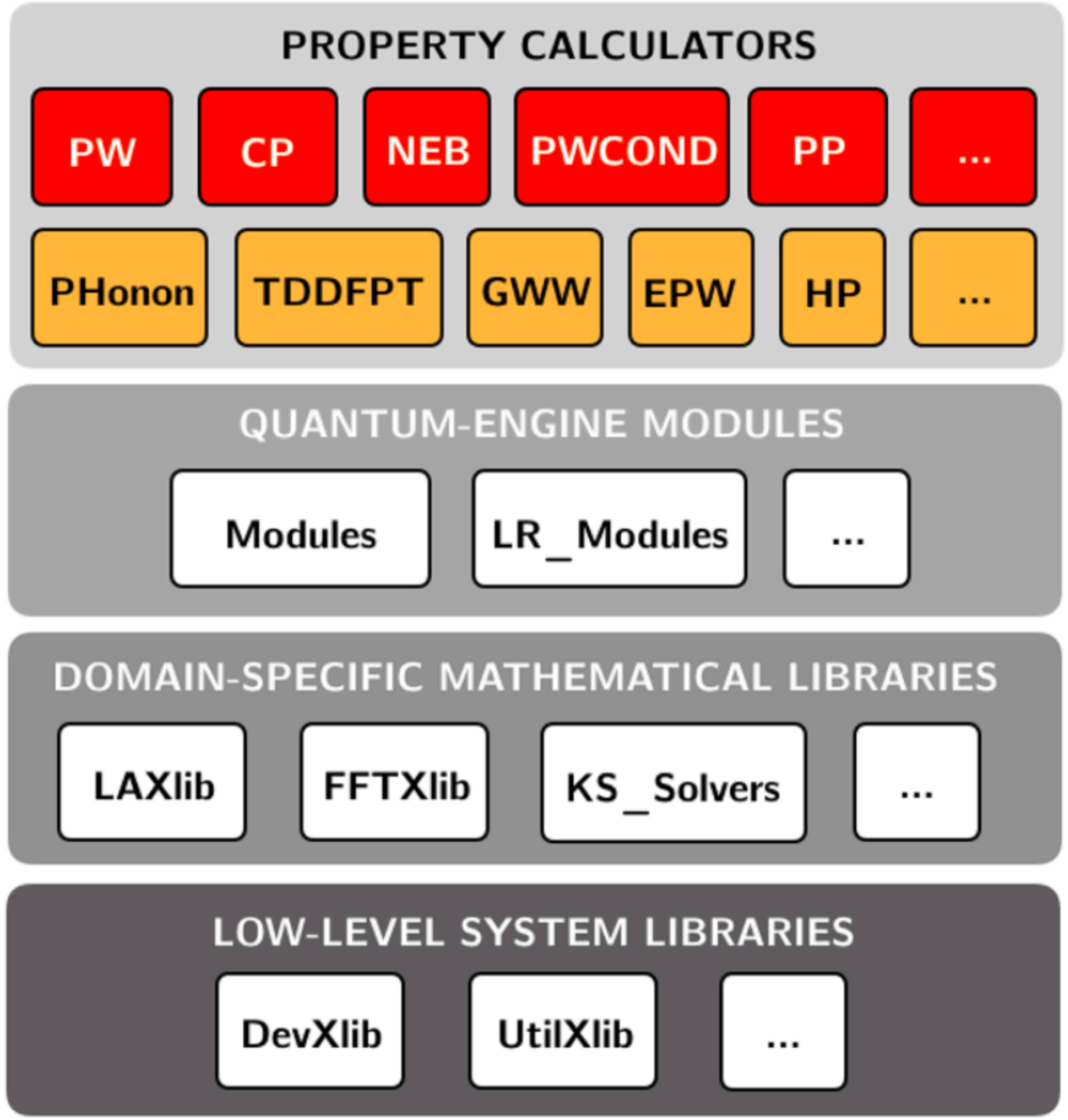
P. Giannozzi, S. Baroni *et al.*, J. Phys.: Condens. Matter 21, 395502 (2009); *ibid.* 29, 465901 (2017).

Quantum ESPRESSO



P. Giannozzi, S. Baroni *et al.*, J. Phys.: Condens. Matter 21, 395502 (2009); *ibid.* 29, 465901 (2017).

Quantum ESPRESSO is a suite of codes

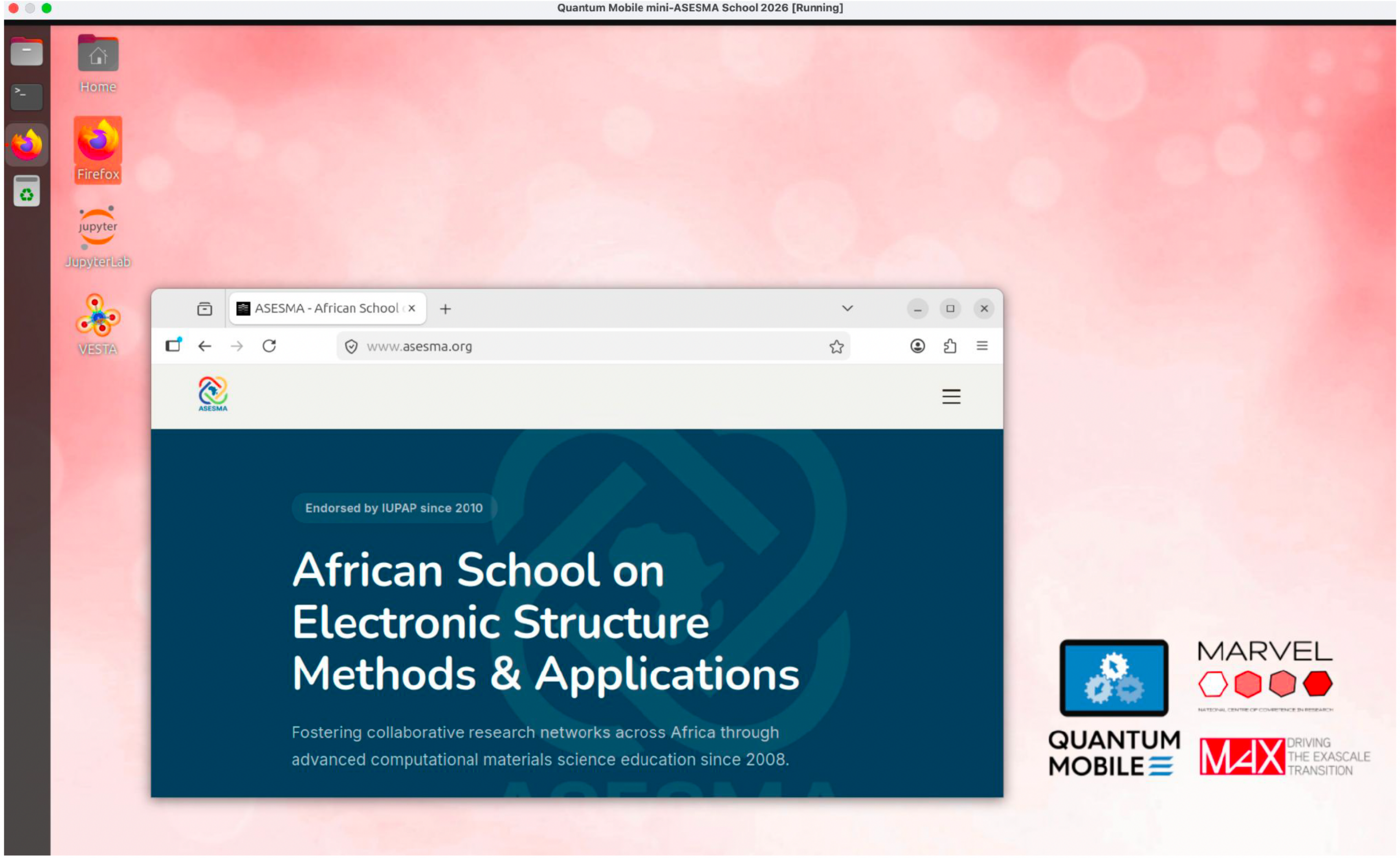


Quantum ESPRESSO is an integrated suite of Open-Source computer codes for electronic-structure calculations and materials modeling at the nanoscale. It is based on density-functional theory, plane waves, and pseudopotentials.

PW - DFT, DFT+U, …
CP - Car-Parrinello molecular dynamics
NEB - Nudged Elastic Bands
PP - Postprocessing tools
Phonon - Phonon calculations
TDDFPT - Spectroscopy
HP - Calculation of Hubbard parameters
…

Quantum Mobile

https://github.com/marvel-nccr/quantum-mobile/blob/support/asesma/custom/README.md



Exercise 1

DFT+*U* for CoO

Check README.md how to run the calculations

```
mpirun -n 2 pw.x < CoO.scf.in |tee CoO.scf.out

mpirun -n 2 pw.x < CoO.nscf.in |tee CoO.nscf.out

mpirun -n 2 projwfc.x < CoO.projwfc.in |tee CoO.projwfc.out

python3 plot_pdos.py
```

Visualize the file CoO_PDOS.pdf

Input file CoO.scf.in

```
&control
   calculation='scf'
   restart_mode='from_scratch',
   prefix='CoO'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
/
&electrons
   conv_thr =  1.d-10
/
ATOMIC_SPECIES
 Co1  58.933194  co_pbesol_v1.2.uspp.F.UPF
 Co2  58.933194  co_pbesol_v1.2.uspp.F.UPF
 O    15.999     O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Co1      0.00000000     0.00000000     0.00000000
Co2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.570726115   0.570726115   1.031099100
   0.570726115   1.031099100   0.570726115
   1.031099100   0.570726115   0.570726115
K_POINTS {automatic}
 3 3 3 0 0 0
```

Self-consistent-field (SCF) calculation

Specification of the lattice (see CELL_PARAMETERS)

Number of atoms and atomic types

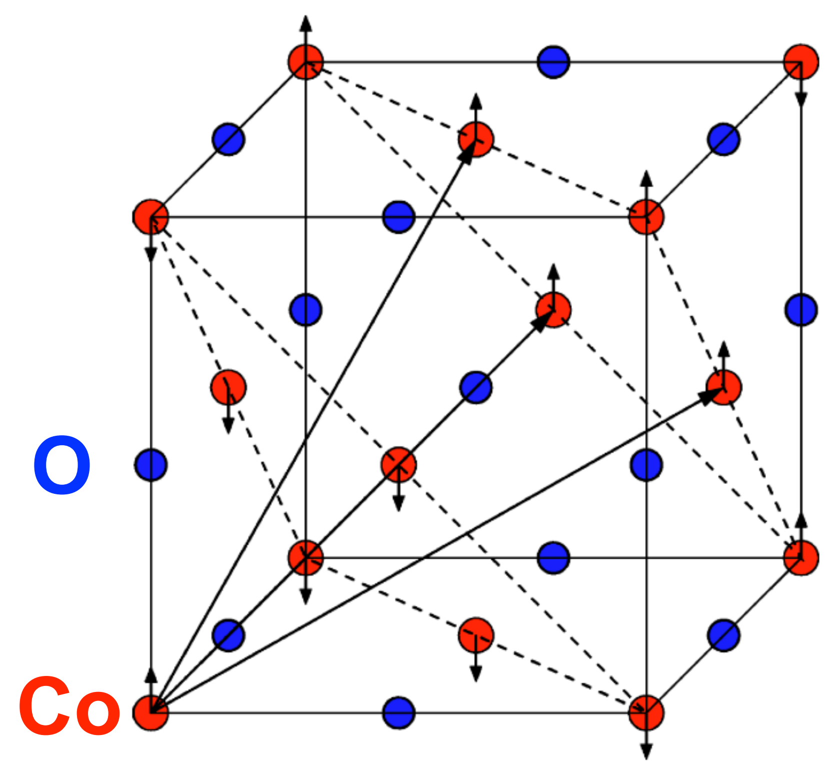
Kinetic-energy cutoff for the wavefunctions and density/potential

Spin-polarized collinear calculation (AFM)

Marzari-Vanderbilt smearing with a broadening parameter of 0.02 Ry

Convergence threshold for self-consistency

Pseudopotentials and functional (PBEsol)



Cell parameters optimized using DFT with the PBEsol functional

Quantum ESPRESSO input generator and structure visualizer

- About the Quantum ESPRESSO input generator and structure visualizer
- Instructions
- Acknowledgements

Upload a crystal structure	Pick an example structure

Upload a file:	Choose file No file chosen
Select the file format:	Quantum ESPRESSO input [parser: qetools]
Select the protocol:[?]	balanced
Select the XC functional:	PBEsol
Select the magnetism/smearing:[?]	non-magnetic metal (fractional occupations)
Refine cell (using spglib):	No

- Advanced settings[?]

By continuing, you agree with the terms of use of this service.

Generate the PWscf input file

https://www.materialscloud.org/work/tools/qeinputgenerator

SSSP pseudopotential library

- pslibrary.1.0.0 US (high acc.)
- pslibrary.1.0.0 PAW (high acc.)
- pslibrary.0.3.1 US (high acc.)
- pslibrary.0.3.1 PAW (high acc.)
- GBRV-1.2 (US)
- GBRV-1.4 (US)
- GBRV-1.5 (US)
- SG15 (NC)
- SG15-1.1 (NC)
- Pseudo Dojo (NC)
- Goedecker
- THEOS
- RE Wentzcovitch (PAW)
- pslibrary.1.0.0 PAW (low acc.)
- Actinides MalteSachs (PAW)

E $\text{WFC}_{\text{(RHO)}}$

E: element
WFC: suggested wave function cutoff in Ry for SSSP efficiency *
(RHO): suggested charge density cutoff in Ry for SSSP efficiency (this is equal to the dual times the wave function cutoff)
*the background color corresponds to the pseudopotential library chosen for SSSP efficiency

SSSP Efficiency (version 1.3.0)

Note: The Ir (Iridium) pseudopotential of the efficiency library has a ghost state at ~10 eV above the Fermi energy, please use it with caution.

Cutoffs table | Pseudos | Switch to SSSP Precision

H $60_{(480)}$																	He $50_{(200)}$
Li $40_{(320)}$	Be $40_{(320)}$											B $35_{(280)}$	C $45_{(360)}$	N $60_{(480)}$	O $50_{(400)}$	F $45_{(360)}$	Ne $50_{(200)}$
Na $40_{(320)}$	Mg $30_{(240)}$											Al $30_{(240)}$	Si $30_{(240)}$	P $30_{(240)}$	S $35_{(280)}$	Cl $40_{(320)}$	Ar $60_{(240)}$
K $60_{(480)}$	Ca $30_{(240)}$	Sc $40_{(160)}$	Ti $35_{(280)}$	V $35_{(280)}$	Cr $40_{(320)}$	Mn $65_{(780)}$	Fe $90_{(1080)}$	Co $45_{(360)}$	Ni $45_{(360)}$	Cu $90_{(720)}$	Zn $40_{(320)}$	Ga $70_{(560)}$	Ge $40_{(320)}$	As $35_{(280)}$	Se $30_{(240)}$	Br $30_{(240)}$	Kr $45_{(180)}$
Rb $30_{(120)}$	Sr $30_{(240)}$	Y $35_{(280)}$	Zr $30_{(240)}$	Nb $40_{(320)}$	Mo $35_{(140)}$	Tc $30_{(120)}$	Ru $35_{(140)}$	Rh $35_{(140)}$	Pd $45_{(180)}$	Ag $50_{(200)}$	Cd $60_{(480)}$	In $50_{(400)}$	Sn $60_{(480)}$	Sb $40_{(320)}$	Te $30_{(240)}$	I $35_{(280)}$	Xe $60_{(240)}$
Cs $30_{(240)}$	Ba $30_{(240)}$	*	Hf $50_{(200)}$	Ta $45_{(360)}$	W $30_{(240)}$	Re $30_{(240)}$	Os $40_{(320)}$	Ir $55_{(440)}$	Pt $35_{(280)}$	Au $45_{(180)}$	Hg $50_{(200)}$	Tl $50_{(400)}$	Pb $40_{(320)}$	Bi $45_{(360)}$	Po $75_{(600)}$	At $80_{(640)}$	Rn $120_{(960)}$
Fr $150_{(1200)}$	Ra $90_{(720)}$	**															
		*	La $40_{(320)}$	Ce $50_{(400)}$	Pr $40_{(320)}$	Nd $40_{(320)}$	Pm $40_{(320)}$	Sm $40_{(320)}$	Eu $40_{(320)}$	Gd $40_{(320)}$	Tb $40_{(320)}$	Dy $40_{(320)}$	Ho $40_{(320)}$	Er $40_{(320)}$	Tm $40_{(320)}$	Yb $40_{(320)}$	Lu $45_{(360)}$
		**	Ac $40_{(320)}$	Th $60_{(480)}$	Pa $60_{(480)}$	U $60_{(480)}$	Np $60_{(480)}$	Pu $60_{(480)}$	Am $60_{(480)}$	Cm $60_{(480)}$	Bk $60_{(480)}$	Cf $60_{(480)}$	Es $60_{(480)}$	Fm $60_{(480)}$	Md $60_{(480)}$	No $60_{(480)}$	Lr $60_{(480)}$

https://www.materialscloud.org/discover/sssp/table/efficiency

Input file CoO.scf.in

```
&control
   calculation='scf'
   restart_mode='from_scratch',
   prefix='CoO'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
/
&electrons
   conv_thr =  1.d-10
/
ATOMIC_SPECIES
 Co1  58.933194  co_pbesol_v1.2.uspp.F.UPF
 Co2  58.933194  co_pbesol_v1.2.uspp.F.UPF
 O    15.999     O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Co1      0.00000000     0.00000000     0.00000000
Co2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.570726115   0.570726115   1.031099100
   0.570726115   1.031099100   0.570726115
   1.031099100   0.570726115   0.570726115
K_POINTS {automatic}
 3 3 3 0 0 0
```

Input file CoO.nscf.in

```
&control
   calculation='nscf'
   restart_mode='from_scratch',
   prefix='CoO'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
   nbnd = 40
/
&electrons
   conv_thr =  1.d-10
/
ATOMIC_SPECIES
 Co1  58.933194  co_pbesol_v1.2.uspp.F.UPF
 Co2  58.933194  co_pbesol_v1.2.uspp.F.UPF
 O    15.999     O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Co1      0.00000000     0.00000000     0.00000000
Co2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.570726115   0.570726115   1.031099100
   0.570726115   1.031099100   0.570726115
   1.031099100   0.570726115   0.570726115
K_POINTS {automatic}
 6 6 6 0 0 0
```

Input file CoO.projwfc.in

```
&projwfc
    prefix='CoO'
    outdir='./tmp'
    ngauss = 0,
    degauss = 0.005,
    Emin = -15.0,
    Emax =  30.0,
    DeltaE = 0.01
 /
```

Gaussian broadening for PDOS

Value of the Gaussian broadening (in Ry)

Minimum and maximum energy for the plot (in eV)

Energy grid step

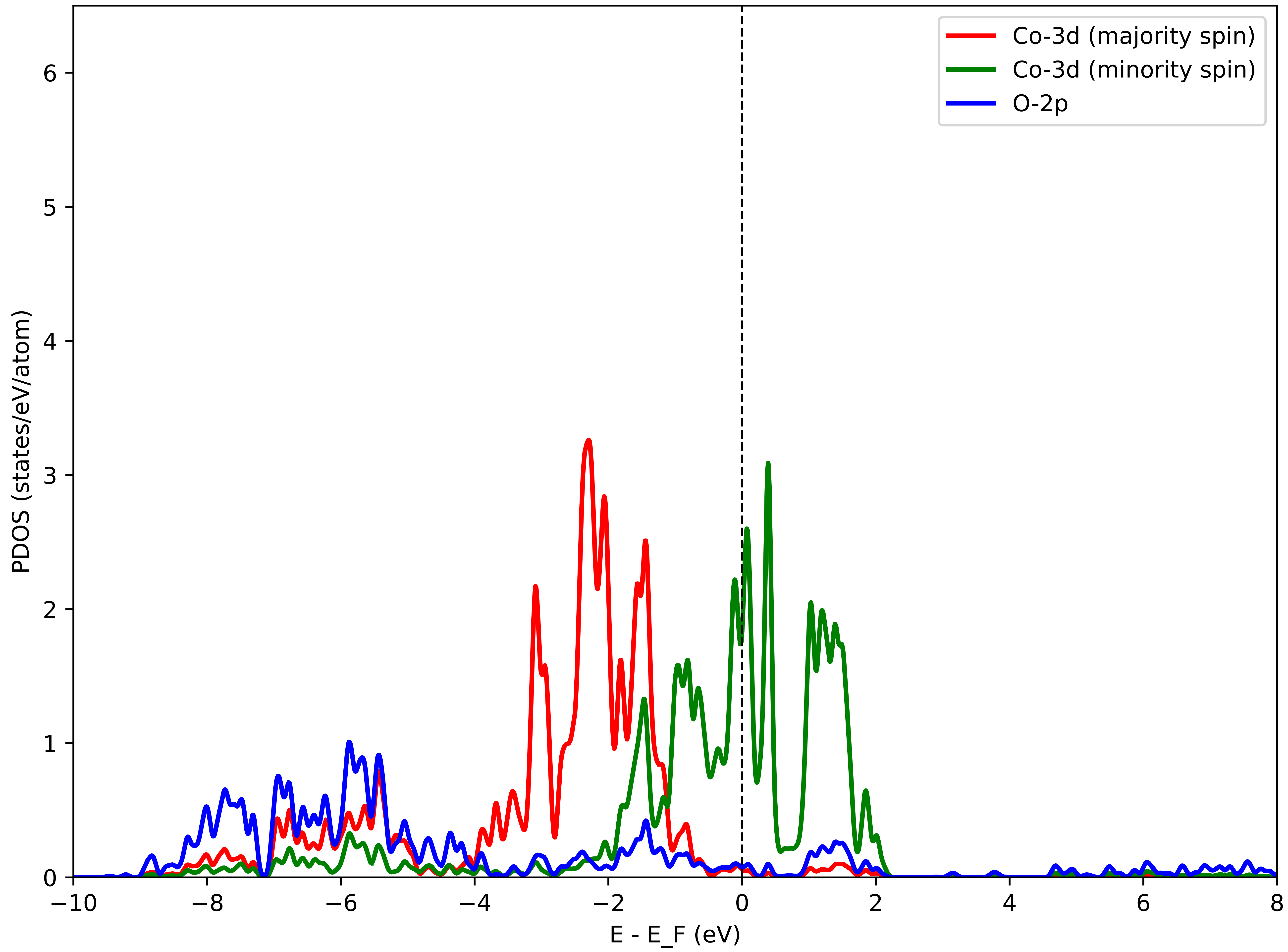
Python3 script: plot_pdos.py

Inspect the script. It aims at plotting Co-3d states (majority spin and minority spin) and O-2p states.

PDOS is shifted such that the Fermi energy corresponds to zero of energy. Use the Fermi energy from the file CoO.scf.out

After running the script, visualize the file CoO_PDOS.pdf

PDOS using DFT (PBEsol)



DFT predicts CoO to be metallic (this is wrong)

Experimentally CoO is insulating

Let's try DFT+U

Input file CoO.scf.in

```
&control
   calculation='scf'
   restart_mode='from_scratch',
   prefix='CoO'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
/
&electrons
   conv_thr =  1.d-10
/
ATOMIC_SPECIES
 Co1   58.933194  co_pbesol_v1.2.uspp.F.UPF
 Co2   58.933194  co_pbesol_v1.2.uspp.F.UPF
 O     15.999     O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Co1      0.00000000     0.00000000     0.00000000
Co2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.570726115   0.570726115   1.031099100
   0.570726115   1.031099100   0.570726115
   1.031099100   0.570726115   0.570726115
K_POINTS {automatic}
 3 3 3 0 0 0
HUBBARD {ortho-atomic}
U Co1-3d 0.0001
U Co2-3d 0.0001
```

HUBBARD card

Name of the card

Type of the Hubbard projectors

```
HUBBARD {ortho-atomic}
U Co1-3d 0.0001
U Co2-3d 0.0001
```

Value of the Hubbard parameter (in eV)

Name of the Hubbard parameter

Hubbard manifold to which we want to apply the Hubbard correction

Atomic type of the chemical element to which we want to apply the Hubbard correction

Detailed description of the new Hubbard input syntax (since v7.1):
See Hubbard_input.pdf

We set some small initial value
just to activate the Hubbard machienary in the code

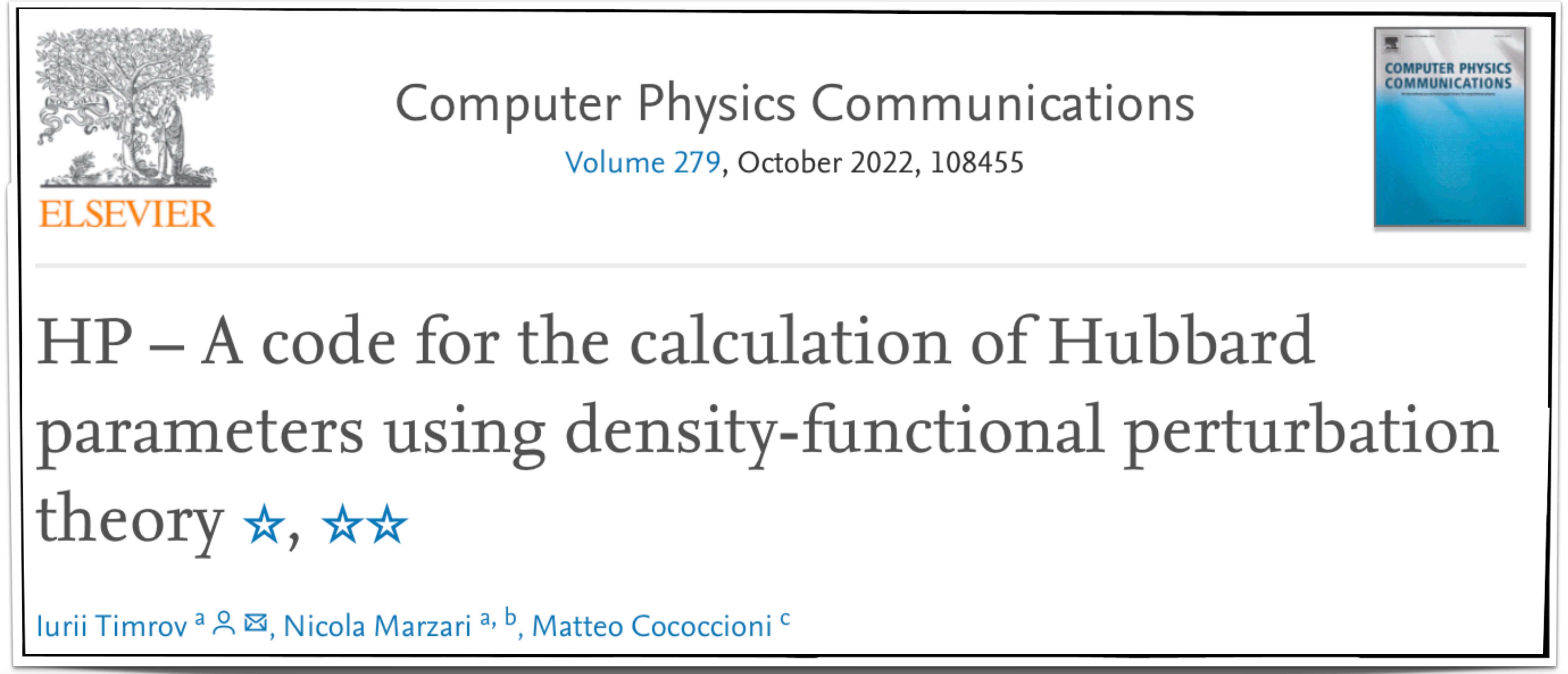
Input file CoO.hp.in

```
&inputhp
    prefix = 'CoO'
    outdir = './tmp'
    nq1 = 2, nq2 = 2, nq3 = 2
    conv_thr_chi = 1.d-6
/
```

The same as in the CoO.scf.in input

Size of the **q** points mesh

Convergence threshold for computing the self-consistent response matrix χ (in eV^{-1})


ELSEVIER

COMPUTER PHYSICS COMMUNICATIONS

Computer Physics Communications

Volume 279, October 2022, 108455

HP – A code for the calculation of Hubbard parameters using density-functional perturbation theory ☆, ☆☆

Iurii Timrov [a], Nicola Marzari [a, b], Matteo Cococcioni [c]

Important notice: The calculation of the Hubbard *U* parameter must be converged (as any other quantity of interest)!

Hubbard *U* must be converged with respect to the kinetic-energy cutoff (ecutwfc and ecutrho), **k** points mesh, and **q** points mesh.

For more details see: *I. Timrov, N. Marzari, M. Cococcioni, Phys. Rev. B* ***98****, 085127 (2018).*

Output file CoO.Hubbard_parameters.dat

```
=-------------------------------------------------------------------=

                    Hubbard U parameters:

   site n.   type  label  spin  new_type  new_label  Hubbard U (eV)
     1         1    Co1     1      1         Co1        6.7553
     2         2    Co2    -1      1         Co1        6.7553

=-------------------------------------------------------------------=
```

These are the output Hubbard *U* parameters for Co1-3d and Co2-3d states

These parameters are computed in a “one-shot” fashion (i.e. from the DFT-PBEsol ground state)

Input file CoO.scf.in

```
&control
   calculation='scf'
   restart_mode='from_scratch',
   prefix='CoO'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
/
&electrons
   conv_thr =  1.d-10
/
ATOMIC_SPECIES
 Co1  58.933194  co_pbesol_v1.2.uspp.F.UPF
 Co2  58.933194  co_pbesol_v1.2.uspp.F.UPF
 O    15.999     O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Co1      0.00000000     0.00000000     0.00000000
Co2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.570726115   0.570726115   1.031099100
   0.570726115   1.031099100   0.570726115
   1.031099100   0.570726115   0.570726115
K_POINTS {automatic}
 3 3 3 0 0 0
HUBBARD {ortho-atomic}
U Co1-3d 6.8
U Co2-3d 6.8
```

HUBBARD card

```
HUBBARD {ortho-atomic}
U Co1-3d 6.8
U Co2-3d 6.8
```

This is the value of the Hubbard *U* parameter (in eV) that we computed using the HP code

We need to specify the same Hubbard *U* in the CoO.nscf.in file

Input file CoO.scf.in

```
&control
   calculation='scf'
   restart_mode='from_scratch',
   prefix='CoO'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
/
&electrons
   conv_thr =  1.d-10
/
ATOMIC_SPECIES
 Co1  58.933194  co_pbesol_v1.2.uspp.F.UPF
 Co2  58.933194  co_pbesol_v1.2.uspp.F.UPF
 O    15.999     O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Co1      0.00000000     0.00000000     0.00000000
Co2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.570726115   0.570726115   1.031099100
   0.570726115   1.031099100   0.570726115
   1.031099100   0.570726115   0.570726115
K_POINTS {automatic}
 3 3 3 0 0 0
HUBBARD {ortho-atomic}
U Co1-3d 6.8
U Co2-3d 6.8
```

Input file CoO.nscf.in

```
&control
   calculation='nscf'
   restart_mode='from_scratch',
   prefix='CoO'
   pseudo_dir = '../../pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
   nbnd = 40
/
&electrons
   conv_thr =  1.d-10
/
ATOMIC_SPECIES
 Co1  58.933194  co_pbesol_v1.2.uspp.F.UPF
 Co2  58.933194  co_pbesol_v1.2.uspp.F.UPF
 O    15.999     O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Co1      0.00000000     0.00000000     0.00000000
Co2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.570726115   0.570726115   1.031099100
   0.570726115   1.031099100   0.570726115
   1.031099100   0.570726115   0.570726115
K_POINTS {automatic}
 6 6 6 0 0 0
HUBBARD {ortho-atomic}
U Co1-3d 6.8
U Co2-3d 6.8
```

Input file CoO.projwfc.in

This is exactly the same as before

```
&projwfc
    prefix='CoO'
    outdir='./tmp'
    ngauss = 0,
    degauss = 0.005,
    Emin = -15.0,
    Emax =  30.0,
    DeltaE = 0.01
 /
```

ngauss: Gaussian broadening for PDOS

degauss: Value of the Gaussian broadening (in Ry)

Emin, Emax: Minimum and maximum energy for the plot (in eV)

DeltaE: Energy grid step

Python3 script: plot_pdos.py

Plot the PDOS using the gnuplot and the plot_pdos.py script

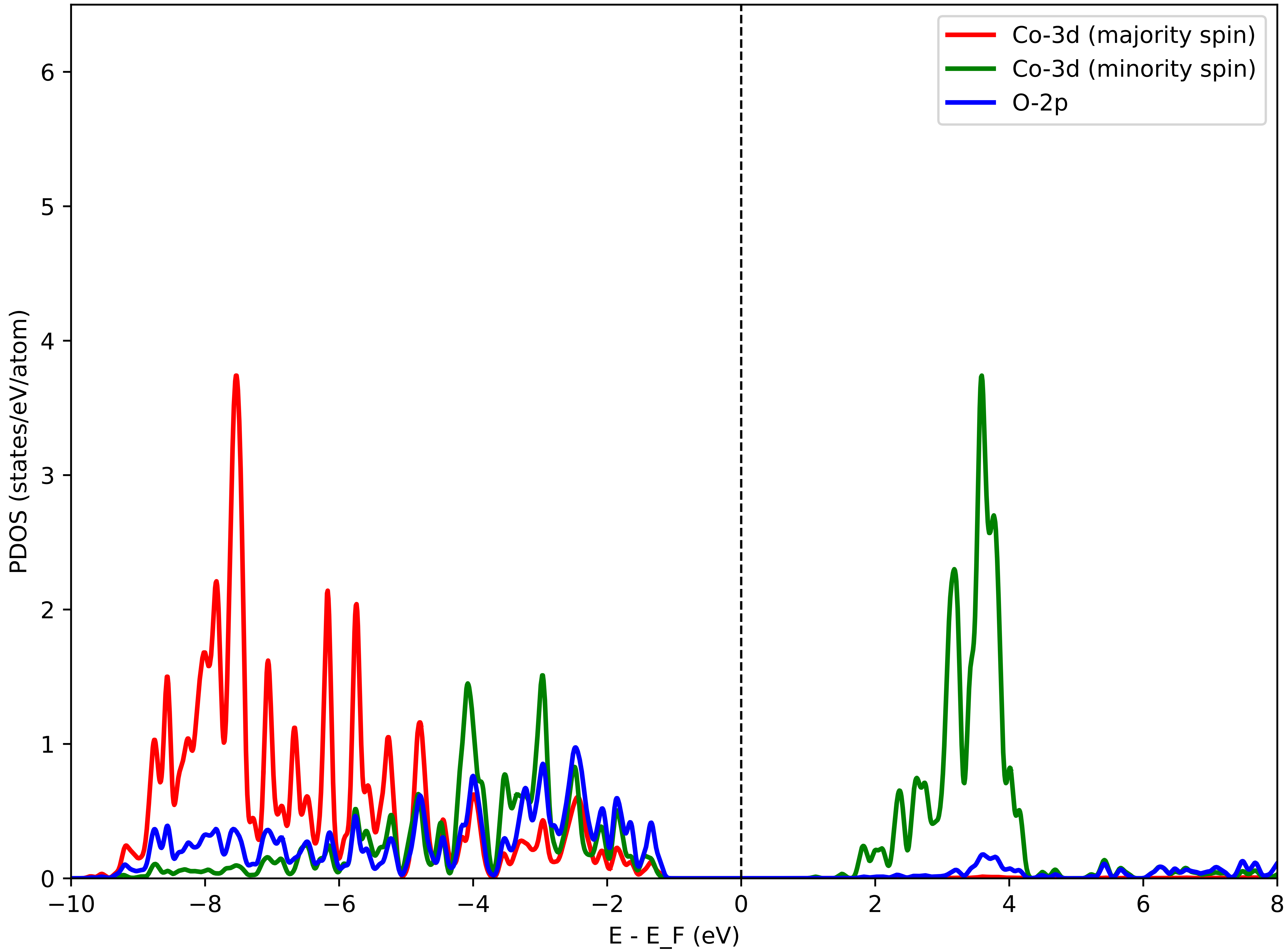
PDOS is shifted such that the Fermi energy corresponds to zero of energy. Use the Fermi energy from the file CoO.scf.out

After running the script, visualize the file CoO_PDOS.pdf

Run the calculations

```
mpirun -n 2 pw.x < CoO.scf.in |tee CoO.scf.out

mpirun -n 2 pw.x < CoO.nscf.in |tee CoO.nscf.out

mpirun -n 2 projwfc.x < CoO.projwfc.in |tee CoO.projwfc.out

python3 plot_pdos.py

Visualize the file CoO_PDOS.pdf
```

PDOS using DFT+*U*

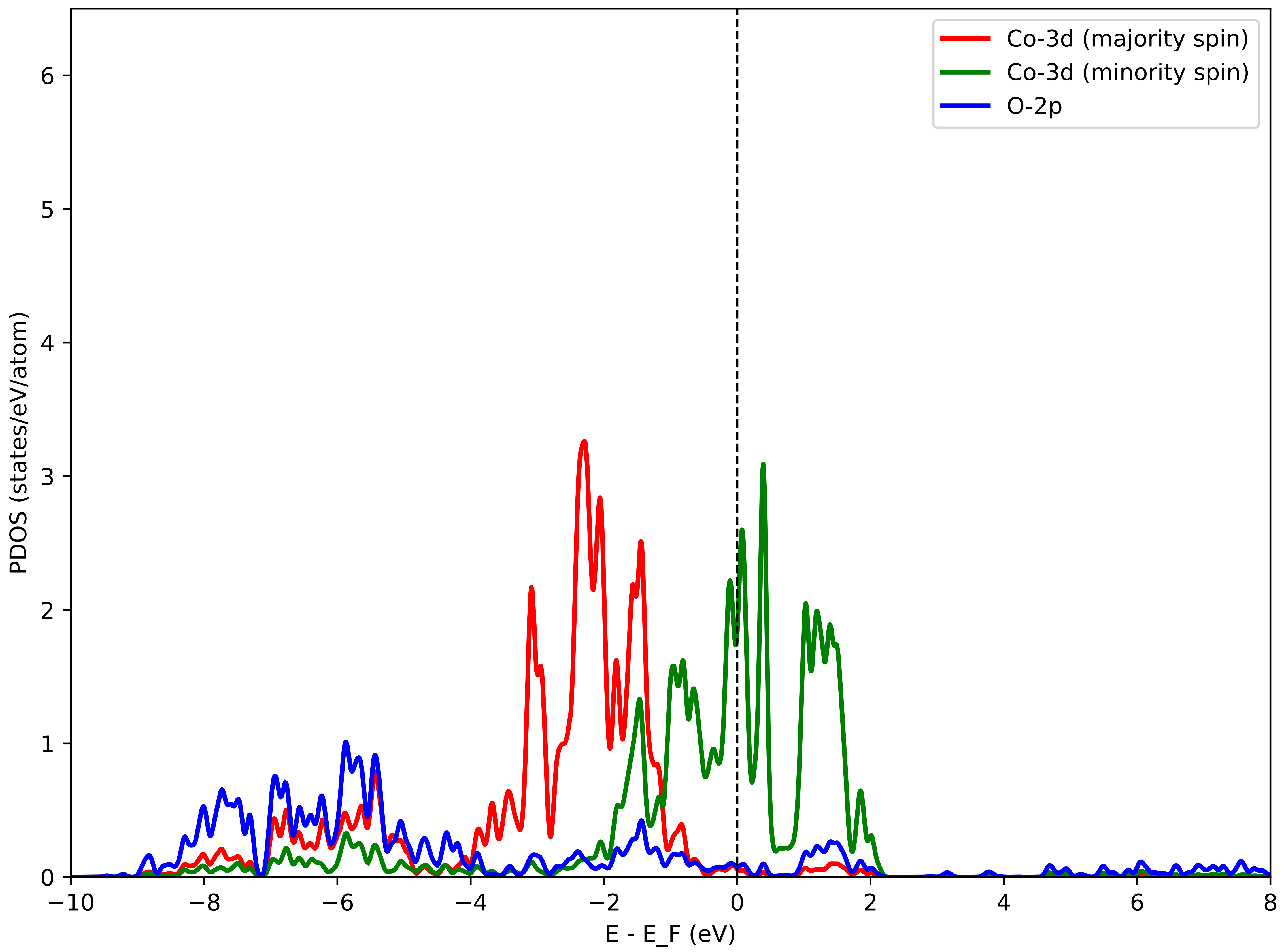


DFT+*U* predicts CoO to be insulating (this is correct)

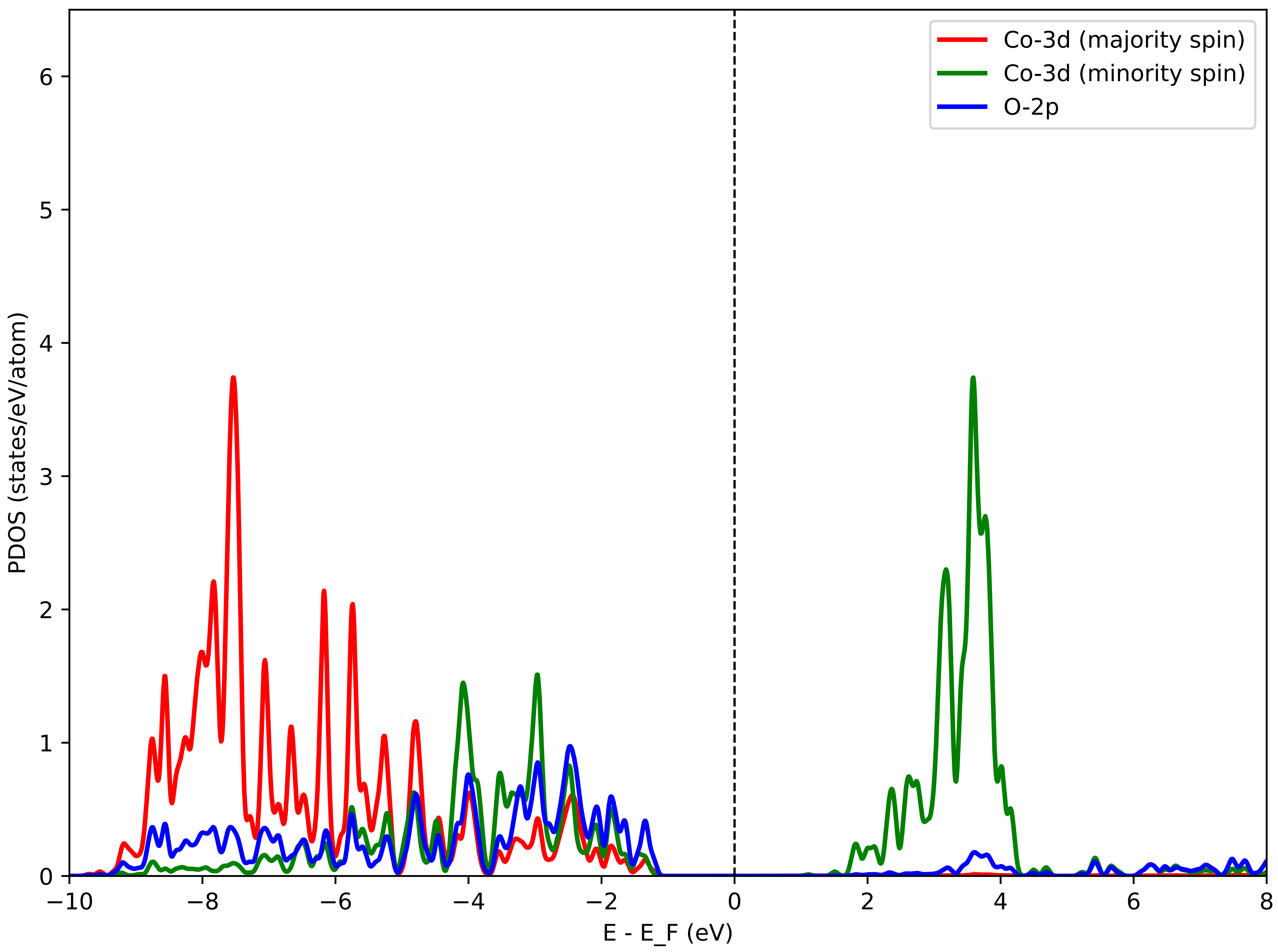
Experimentally CoO is insulating

Agreement with experiment

PDOS using DFT



PDOS using DFT+U



Co-3d (minority spin) states are split

Co-3d (majority spin) states are shifted to lower energies

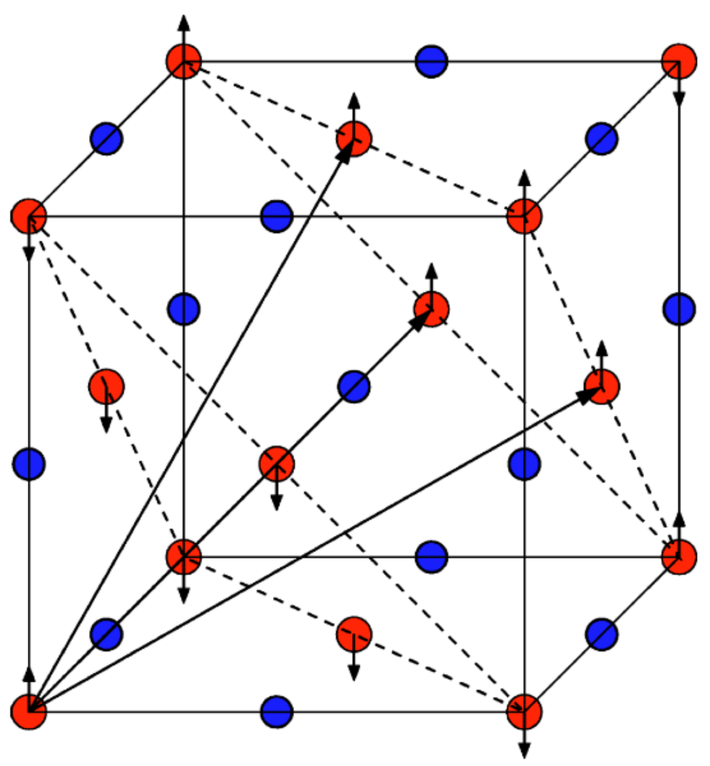
DFT+U band gap: 2.4 eV Experimental gap: 2.5±0.3 eV

Exercise 2

Projected density of states of FeO with and without the Hubbard *U* correction

Input file FeO.scf.in

```
&control
    calculation='scf'
    restart_mode='from_scratch',
    prefix='FeO'
    pseudo_dir = '../../files/pseudo'
    outdir='./tmp'
    verbosity='high'
 /
 &system
    ibrav = 0,
    celldm(1) = 8.00,
    nat = 4,
    ntyp = 3,
    ecutwfc = 35.0
    ecutrho = 280.0
    nspin = 2
    starting_magnetization(1) =  0.5,
    starting_magnetization(2) = -0.5,
    occupations = 'smearing',
    smearing = 'mv',
    degauss = 0.02
 /
 &electrons
    conv_thr =  1.d-10
 /
ATOMIC_SPECIES
 Fe1  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 Fe2  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 O    15.999  O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Fe1      0.00000000     0.00000000     0.00000000
Fe2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.582729109   0.582729109   1.045799900
   0.582729109   1.045799900   0.582729109
   1.045799900   0.582729109   0.582729109
K_POINTS {automatic}
 3 3 3 0 0 0
HUBBARD {ortho-atomic}
U Fe1-3d 0.0001
U Fe2-3d 0.0001
```



Input file FeO.nscf.in

```
&control
   calculation='nscf'
   restart_mode='from_scratch',
   prefix='FeO'
   pseudo_dir = '../files/pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
   nbnd = 40
/
&electrons
   conv_thr =  1.d-9
/
ATOMIC_SPECIES
 Fe1  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 Fe2  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 O    15.999  O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Fe1      0.00000000     0.00000000     0.00000000
Fe2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.582729109   0.582729109   1.045799900
   0.582729109   1.045799900   0.582729109
   1.045799900   0.582729109   0.582729109
K_POINTS {automatic}
 6 6 6 0 0 0
HUBBARD {ortho-atomic}
U Fe1-3d 0.0001
U Fe2-3d 0.0001
```

Input file FeO.projwfc.in

```
&projwfc
    prefix='FeO'
    outdir='./tmp'
    ngauss = 0,          Gaussian broadening for PDOS
    degauss = 0.005,     Value of the Gaussian broadening (in Ry)
    Emin = -15.0,
    Emax =  30.0,        Minimum and maximum energy for the plot (in eV)
    DeltaE = 0.01        Energy grid step
 /
```

Gnuplot script: plot_pdos.gp

Inspect the script. It aims at plotting Fe-3d states (majority spin and minority spin) and O-2p states.

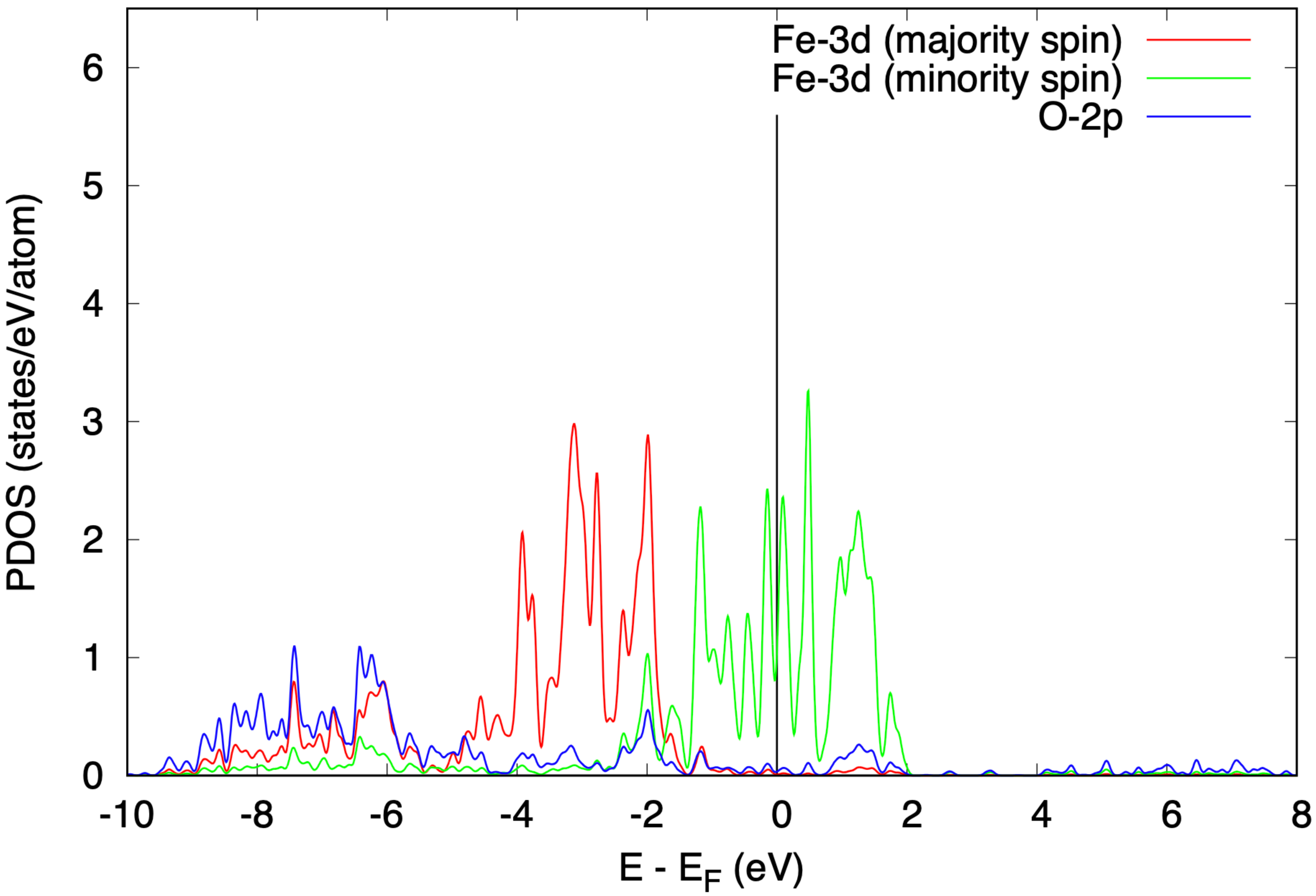
PDOS is shifted such that the Fermi energy corresponds to zero of energy.

After running the script, visualize the file FeO_PDOS.eps.

Run the calculations

```
pw.x < FeO.scf.in |tee FeO.scf.out

pw.x < FeO.nscf.in |tee FeO.nscf.out

projwfc.x < FeO.projwfc.in |tee FeO.projwfc.out

gnuplot plot_pdos.gp

evince FeO_PDOS.eps
```

PDOS using DFT (PBEsol)



DFT predicts FeO to be metallic (this is wrong)

Experimentally FeO is insulating

Let's try DFT+U

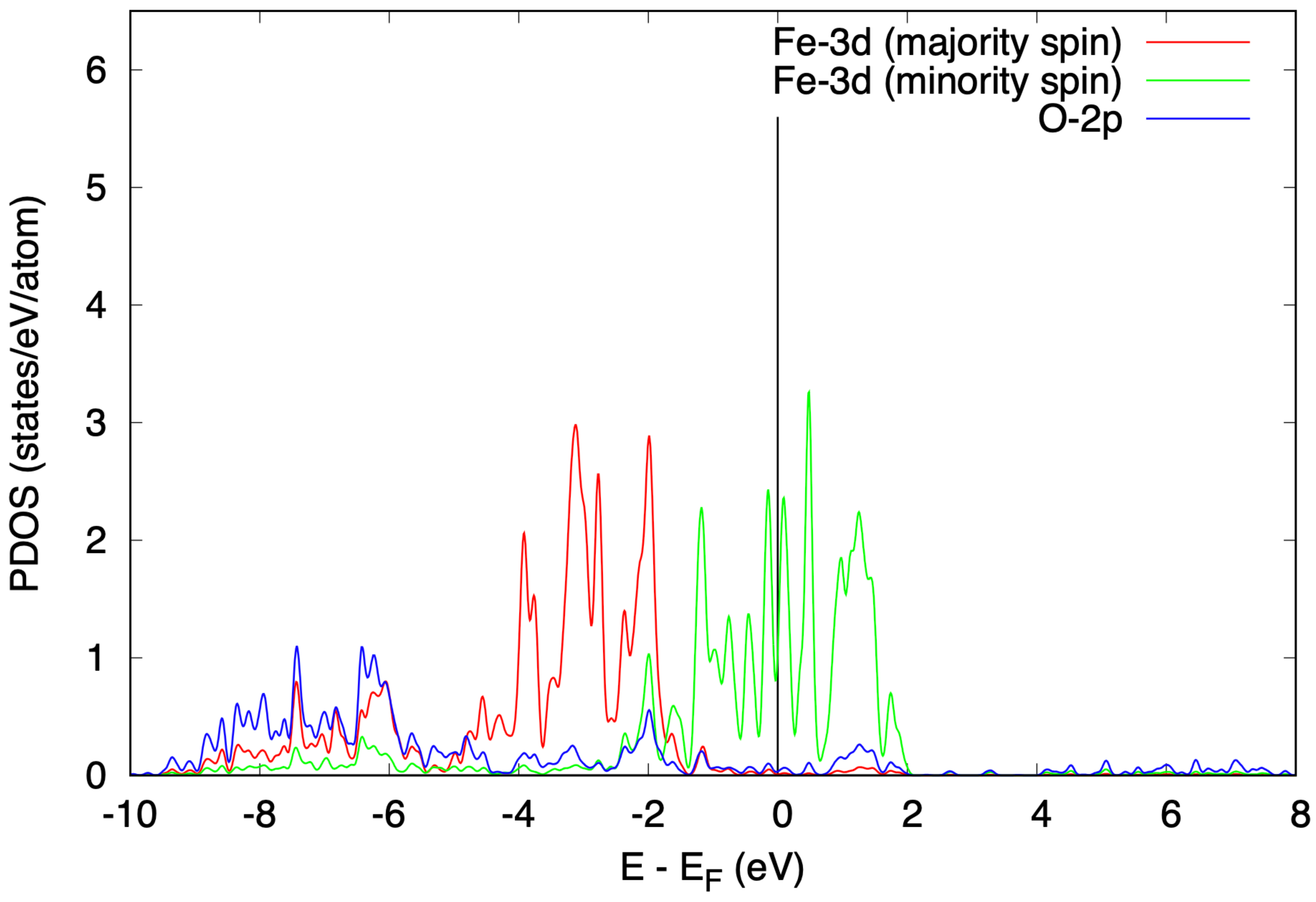
Input file FeO.scf.in

```
&control
    calculation='scf'
    restart_mode='from_scratch',
    prefix='FeO'
    pseudo_dir = '../../files/pseudo'
    outdir='./tmp'
    verbosity='high'
 /
 &system
    ibrav = 0,
    celldm(1) = 8.00,
    nat = 4,
    ntyp = 3,
    ecutwfc = 35.0
    ecutrho = 280.0
    nspin = 2
    starting_magnetization(1) =  0.5,
    starting_magnetization(2) = -0.5,
    occupations = 'smearing',
    smearing = 'mv',
    degauss = 0.02
 /
 &electrons
    conv_thr =  1.d-10
 /
ATOMIC_SPECIES
 Fe1  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 Fe2  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 O    15.999  O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Fe1      0.00000000     0.00000000     0.00000000
Fe2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.582729109   0.582729109   1.045799900
   0.582729109   1.045799900   0.582729109
   1.045799900   0.582729109   0.582729109
K_POINTS {automatic}
 3 3 3 0 0 0
HUBBARD {ortho-atomic}
U Fe1-3d 5.36
U Fe2-3d 5.36
```

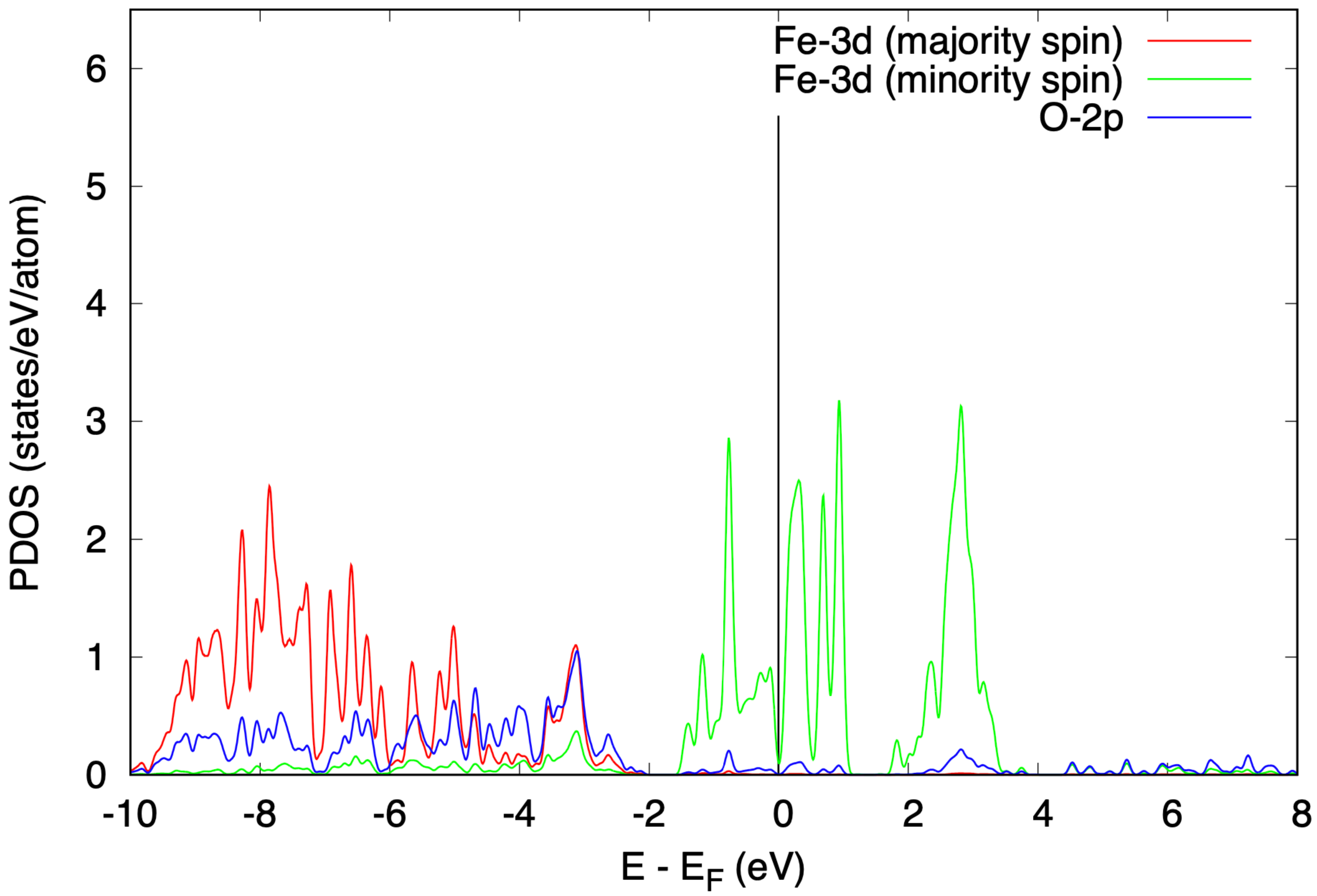
Input file FeO.nscf.in

```
&control
   calculation='nscf'
   restart_mode='from_scratch',
   prefix='FeO'
   pseudo_dir = '../files/pseudo'
   outdir='./tmp'
   verbosity='high'
/
&system
   ibrav = 0,
   celldm(1) = 8.00,
   nat = 4,
   ntyp = 3,
   ecutwfc = 35.0
   ecutrho = 280.0
   nspin = 2
   starting_magnetization(1) =  0.5,
   starting_magnetization(2) = -0.5,
   occupations = 'smearing',
   smearing = 'mv',
   degauss = 0.02
   nbnd = 40
/
&electrons
   conv_thr =  1.d-9
/
ATOMIC_SPECIES
 Fe1  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 Fe2  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 O    15.999  O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Fe1      0.00000000     0.00000000     0.00000000
Fe2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.582729109   0.582729109   1.045799900
   0.582729109   1.045799900   0.582729109
   1.045799900   0.582729109   0.582729109
K_POINTS {automatic}
 6 6 6 0 0 0
HUBBARD {ortho-atomic}
U Fe1-3d 5.36
U Fe2-3d 5.36
```

PDOS using DFT



PDOS using DFT+*U*



Even a standard DFT+*U* calculation does not manage to open a gap in FeO

Output file FeO.scf.out from the DFT+*U* calculation

After the 1st SCF iteration:

```
================== HUBBARD OCCUPATIONS ===================
------------------------ ATOM    1 ------------------------
Tr[ns(  1)] (up, down, total) =   4.98138  0.94209  5.92348
Atomic magnetic moment for atom   1 =   4.03929
SPIN  1
eigenvalues:
  0.996  0.996  0.996  0.997  0.997
SPIN  2
eigenvalues:
  0.075  0.075  0.253  0.253  0.285
------------------------ ATOM    2 ------------------------
Tr[ns(  2)] (up, down, total) =   0.94168  4.98133  5.92301
Atomic magnetic moment for atom   2 =  -4.03965
SPIN  1
eigenvalues:
  0.075  0.075  0.253  0.253  0.285
SPIN  2
eigenvalues:
  0.996  0.996  0.996  0.997  0.997
```

Output file FeO.scf.out from the DFT+*U* calculation

After the 1st SCF iteration:

```
=================== HUBBARD OCCUPATIONS ===================
------------------------ ATOM    1 ------------------------
Tr[ns(  1)] (up, down, total) =   4.98138  0.94209  5.92348
Atomic magnetic moment for atom   1 =   4.03929
SPIN  1
eigenvalues:
  0.996  0.996  0.996  0.997  0.997
SPIN  2
eigenvalues:
  0.075  0.075  0.253  0.253  0.285
------------------------ ATOM    2 ------------------------
Tr[ns(  2)] (up, down, total) =   0.94168  4.98133  5.92301
Atomic magnetic moment for atom   2 =  -4.03965
SPIN  1
eigenvalues:
  0.075  0.075  0.253  0.253  0.285
SPIN  2
eigenvalues:
  0.996  0.996  0.996  0.997  0.997
```

starting_ns_eigenvalue(5,2,1) = 1.0

starting_ns_eigenvalue(5,1,2) = 1.0

Output file FeO.scf.out from the DFT+*U* calculation

After the 1st SCF iteration:

```
=================== HUBBARD OCCUPATIONS ===================
------------------------ ATOM    1 ------------------------
Tr[ns(  1)] (up, down, total) =   4.98138  0.94209  5.92348
Atomic magnetic moment for atom   1 =    4.03929
SPIN  1
eigenvalues:
  0.996  0.996  0.996  0.997  0.997
SPIN  2
eigenvalues:
  0.075  0.075  0.253  0.253  0.285
------------------------ ATOM    2 ------------------------
Tr[ns(  2)] (up, down, total) =   0.94168  4.98133  5.92301
Atomic magnetic moment for atom   2 =  -4.03965
SPIN  1
eigenvalues:
  0.075  0.075  0.253  0.253  0.285
SPIN  2
eigenvalues:
  0.996  0.996  0.996  0.997  0.997
```

starting_ns_eigenvalue(5,2,1) = 1.0

5th eigenvalue

spin index

atomic type

starting_ns_eigenvalue(5,1,2) = 1.0

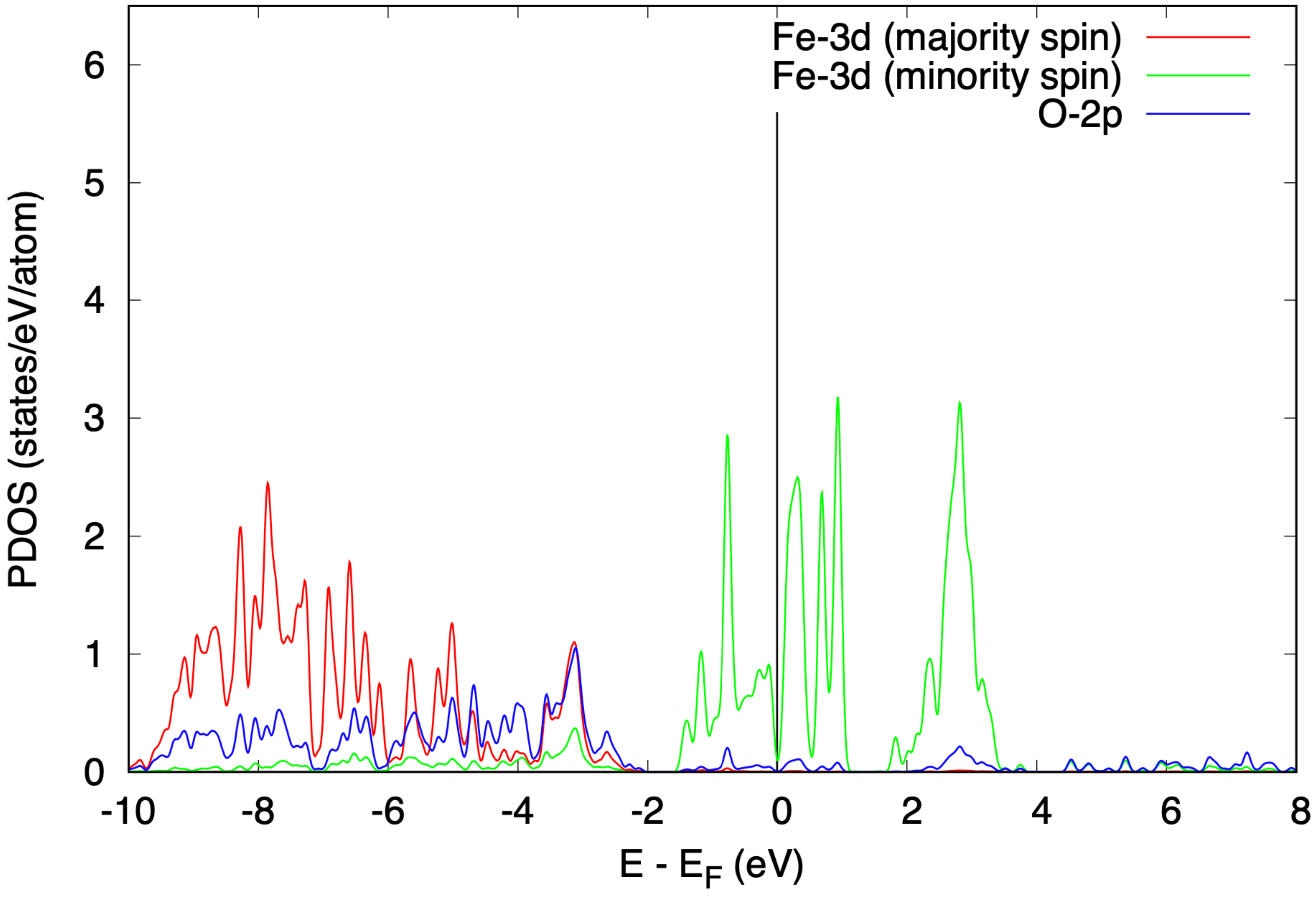
Input file FeO.scf.in

```
 &control
    calculation='scf'
    restart_mode='from_scratch',
    prefix='FeO'
    pseudo_dir = '../../files/pseudo'
    outdir='./tmp'
    verbosity='high'
 /
 &system
    ibrav = 0,
    celldm(1) = 8.00,
    nat = 4,
    ntyp = 3,
    ecutwfc = 35.0
    ecutrho = 280.0
    nspin = 2
    starting_magnetization(1) =  0.5,
    starting_magnetization(2) = -0.5,
    occupations = 'smearing',
    smearing = 'mv',
    degauss = 0.02
    starting_ns_eigenvalue(5,2,1) = 1.0
    starting_ns_eigenvalue(5,1,2) = 1.0
 /
 &electrons
    conv_thr =  1.d-10
 /
ATOMIC_SPECIES
 Fe1  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 Fe2  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 O    15.999  O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Fe1      0.00000000     0.00000000     0.00000000
Fe2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.582729109   0.582729109   1.045799900
   0.582729109   1.045799900   0.582729109
   1.045799900   0.582729109   0.582729109
K_POINTS {automatic}
 3 3 3 0 0 0
HUBBARD {ortho-atomic}
U Fe1-3d 5.36
U Fe2-3d 5.36
```

Input file FeO.nscf.in

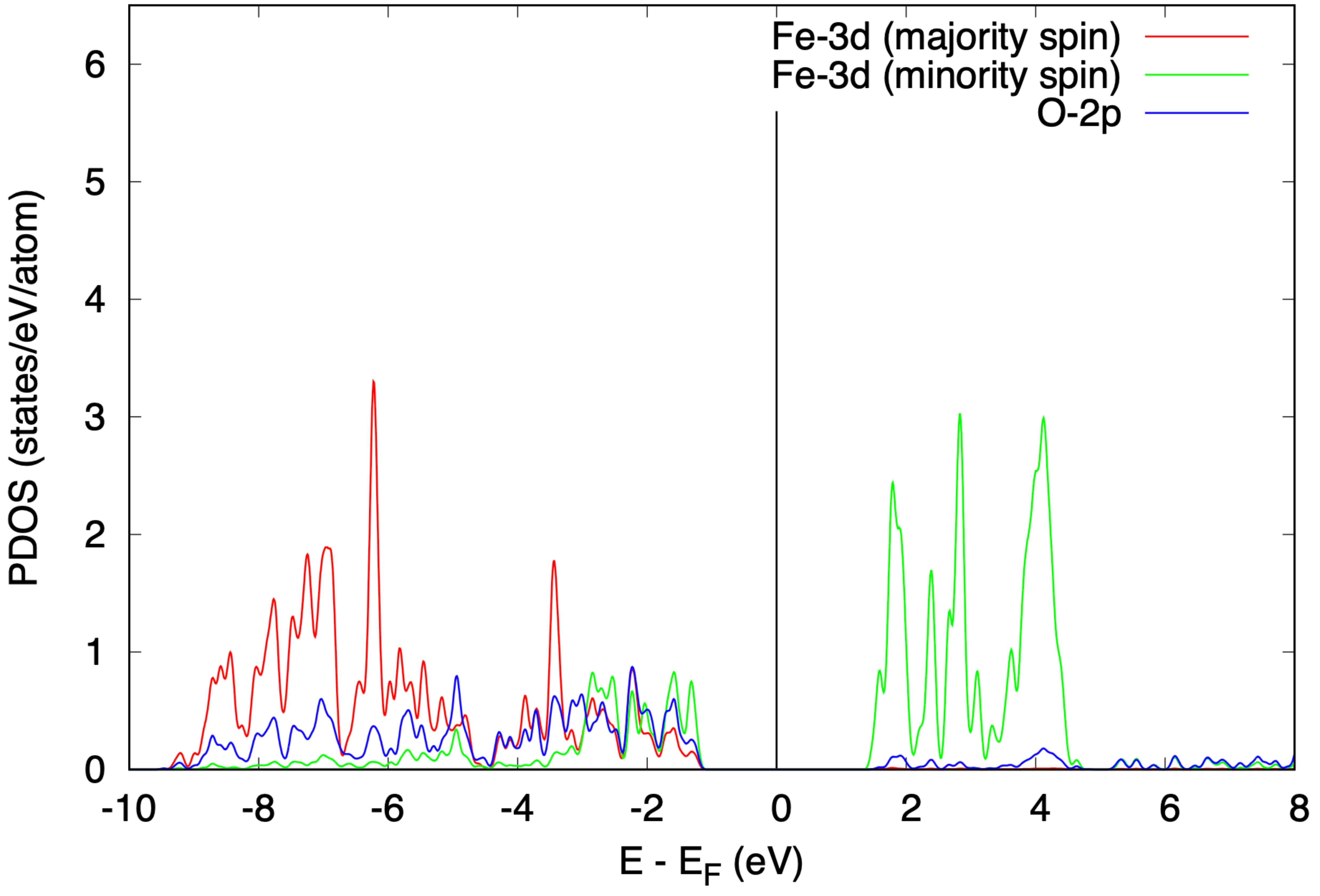
```
 &control
    calculation='nscf'
    restart_mode='from_scratch',
    prefix='FeO'
    pseudo_dir = '../files/pseudo'
    outdir='./tmp'
    verbosity='high'
 /
 &system
    ibrav = 0,
    celldm(1) = 8.00,
    nat = 4,
    ntyp = 3,
    ecutwfc = 35.0
    ecutrho = 280.0
    nspin = 2
    starting_magnetization(1) =  0.5,
    starting_magnetization(2) = -0.5,
    occupations = 'smearing',
    smearing = 'mv',
    degauss = 0.02
    starting_ns_eigenvalue(5,2,1) = 1.0
    starting_ns_eigenvalue(5,1,2) = 1.0
    nbnd = 40
 /
 &electrons
    conv_thr =  1.d-9
 /
ATOMIC_SPECIES
 Fe1  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 Fe2  55.845  Fe.pbesol-spn-kjpaw_psl.0.2.1.UPF
 O    15.999  O.pbesol-n-kjpaw_psl.0.1.UPF
ATOMIC_POSITIONS {crystal}
Fe1      0.00000000     0.00000000     0.00000000
Fe2      0.50000000     0.50000000     0.50000000
O        0.25000000     0.25000000     0.25000000
O        0.75000000     0.75000000     0.75000000
CELL_PARAMETERS {alat}
   0.582729109   0.582729109   1.045799900
   0.582729109   1.045799900   0.582729109
   1.045799900   0.582729109   0.582729109
K_POINTS {automatic}
 6 6 6 0 0 0
HUBBARD {ortho-atomic}
U Fe1-3d 5.36
U Fe2-3d 5.36
```

PDOS using DFT+*U*



E_{TOT} = -735.372 Ry

PDOS using DFT+*U* (with starting_ns_eigenvalue)



E_{TOT} = -735.480 Ry

DFT+*U* with starting_ns_eigenvalue opens the band gap in FeO

DFT+*U* band gap: 2.76 eV Experimental gap: 2.4 eV